

Local Structural Channels in SAT

A Diagnostic Analysis of the First External Gate Challenge

MAAT Structural Selection Series – Paper 70

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Abstract

Paper 69 executed the preregistered MAAT v1.7 Gate Challenge protocol on a family-balanced public SATLIB smoke subset. The external pipeline passed manifest validation and SHA256 verification, but the frozen root-level gate did not receive positive support and lost clearly against the classical MOMS heuristic. This paper does not modify the gate, introduce a new MAAT version, or propose a new SAT heuristic. Instead, it asks a diagnostic question: *which local SAT information channels does MOMS exploit that the frozen gate does not see?*

Using the same 30 SATLIB smoke instances, we compute post-hoc local structural channels from the CNF formulas: shortest-clause pressure, shortest-clause variable entropy, variable-occurrence dispersion, Jeroslow–Wang concentration, clause-length entropy, and unit-propagation cascade proxies. The target is the Paper-69 gate-to-MOMS regret, where positive regret means that the frozen gate used more compute than MOMS.

The strongest diagnostic signal is not the root gate itself ($\rho_S = -0.176$), but local short-clause and occurrence geometry. Shortest-clause variable entropy correlates positively with gate-to-MOMS regret ($\rho_S = 0.643$), while degree/occurrence concentration correlates negatively ($\rho_S = -0.567$ for the Paper-69 degree coefficient of variation and $\rho_S = -0.488$ for variable-occurrence coefficient of variation). Family-level regret is largest on Dubois and pigeonhole instances and nearly zero on the graph-colouring and small random-3SAT families.

The conclusion is deliberately narrow. The first external SAT Gate Challenge did not fail because the computation was irreproducible or because the data pipeline was weak. It failed because the frozen root-level gate lacks a decision-local short-clause/occurrence channel that classical SAT heuristics exploit directly. This identifies missing structural information; it does not justify retuning Paper 69 retroactively.

1 Purpose

Paper 69 reported the first external SAT/CDCL smoke execution of the Paper-68 Gate Challenge protocol. The result was methodologically successful but not confirmatory:

$$\Delta U(\text{gate} - \text{score-}R_{\text{rob}}) = 7.292, \quad 95\% \text{ CI} = [-9.493, 24.509],$$

while against MOMS the gate lost clearly:

$$\Delta U(\text{gate} - \text{MOMS}) = -66.392, \quad 95\% \text{ CI} = [-107.679, -28.770].$$

The purpose of Paper 70 is not to rescue the gate. It is to understand the negative result. The central question is:

Which local SAT channels are present in the public SATLIB smoke subset but absent from the frozen MAAT v1.7 root-level gate?

This distinction matters. Changing the gate after seeing Paper-69 outcomes would violate the spirit of the preregistered Gate Challenge. Diagnosing what information was missing is different. It converts a negative external result into a testable account of model incompleteness.

2 Source Experiment

The analysis uses the Paper-69 SATLIB smoke execution:

- 30 public SATLIB instances,
- six benchmark families,
- five instances per family,
- conflict budget 1000,
- policies VSIDS, MOMS, Jeroslow–Wang, progress-only, score-with- R_{rob} , gate-v1.7, and shuffled- R_{rob} gate null.

The Paper-68 protocol was preregistered before external Gate Challenge execution:

<https://doi.org/10.5281/zenodo.20882852>.

Raw SATLIB CNF files are used only as local inputs for the diagnostic script. They are not redistributed. Published outputs contain only derived local-channel features, correlations, figures, and summaries.

3 Diagnostic Target

For each instance i , define the gate-to-MOMS regret as

$$r_i = C_i(\text{gate-v1.7}) - C_i(\text{MOMS}), \quad (1)$$

where C_i is the Paper-69 compute cost. In the output files this is reported as

regret_gate_to_moms.

Positive r_i means the gate used more compute than MOMS. Negative r_i would mean the gate beat MOMS.

The analysis then computes Spearman and Pearson correlations between r_i and candidate local structural channels. The main reported statistic is Spearman correlation, because the smoke subset is small and family-structured.

4 Local Structural Channels

Let a CNF formula have n variables and m clauses $\mathcal{C} = \{C_j\}_{j=1}^m$. Let

$$\ell_j = |C_j|, \quad \ell_{\min} = \min_j \ell_j, \quad \mathcal{C}_{\min} = \{C_j : \ell_j = \ell_{\min}\}.$$

4.1 Short-clause pressure

MOMS is explicitly sensitive to variables appearing in shortest clauses. The simplest pressure channel is

$$P_{\min} = \frac{|\mathcal{C}_{\min}|}{n}. \quad (2)$$

This measures how many shortest clauses press on the variable set.

4.2 Shortest-clause variable geometry

Let $a_v^{(\min)}$ be the number of occurrences of variable v in shortest clauses. Two derived quantities are used:

$$c_{\min} = \frac{\max_v a_v^{(\min)}}{\sum_u a_u^{(\min)} + \epsilon}, \quad (3)$$

$$H_{\min} = -\frac{1}{\log k} \sum_{v: a_v^{(\min)} > 0} p_v \log(p_v + \epsilon), \quad p_v = \frac{a_v^{(\min)}}{\sum_u a_u^{(\min)}}. \quad (4)$$

Here $H_{\min}$ is the normalized entropy of shortest-clause variable pressure. High $H_{\min}$ means the shortest-clause pressure is spread broadly across many variables; low $H_{\min}$ means it is concentrated.

The MOMS-local signal used for this diagnostic is

$$P_{\text{MOMS}} = P_{\min}(1 + c_{\min}). \quad (5)$$

This is not a replacement for MOMS. It is a compact diagnostic proxy for shortest-clause pressure plus local variable concentration.

4.3 Occurrence and weighted-clause channels

The analysis also computes:

- variable-occurrence coefficient of variation,
- polarity imbalance,
- clause-length entropy,
- Jeroslow–Wang variable concentration,
- weighted clause pressure $\langle 2^{-|C|} \rangle$,
- unit-propagation cascade proxies after single-literal assumptions.

These channels are intentionally local. They are closer to decision-level SAT heuristics than to the root-level R_{rob} gate used in Paper 69.

5 Results

5.1 Channel correlations

Table 1 shows the strongest local-channel correlations with gate-to-MOMS regret. The root-level gate itself is weakly correlated with the regret target:

$$\rho_S(G_{\text{gate}}, r) = -0.176.$$

By contrast, local short-clause and occurrence-geometry channels are stronger.

Channel	Spearman ρ_S	Pearson r
Shortest-clause variable entropy	0.643	0.538
Paper-69 degree CV	−0.567	−0.530
Variable-occurrence CV	−0.488	−0.638
Short-clause pressure	−0.389	−0.134
MOMS-local signal	−0.338	−0.146
Literal polarity imbalance	−0.265	−0.247
Jeroslow–Wang concentration	−0.217	−0.259
Gate value G_{gate}	−0.176	−0.167

Table 1: Local structural channels versus Paper-69 gate-to-MOMS regret. Positive regret means the frozen gate used more compute than MOMS. The strongest signal is not the root gate but shortest-clause variable geometry.

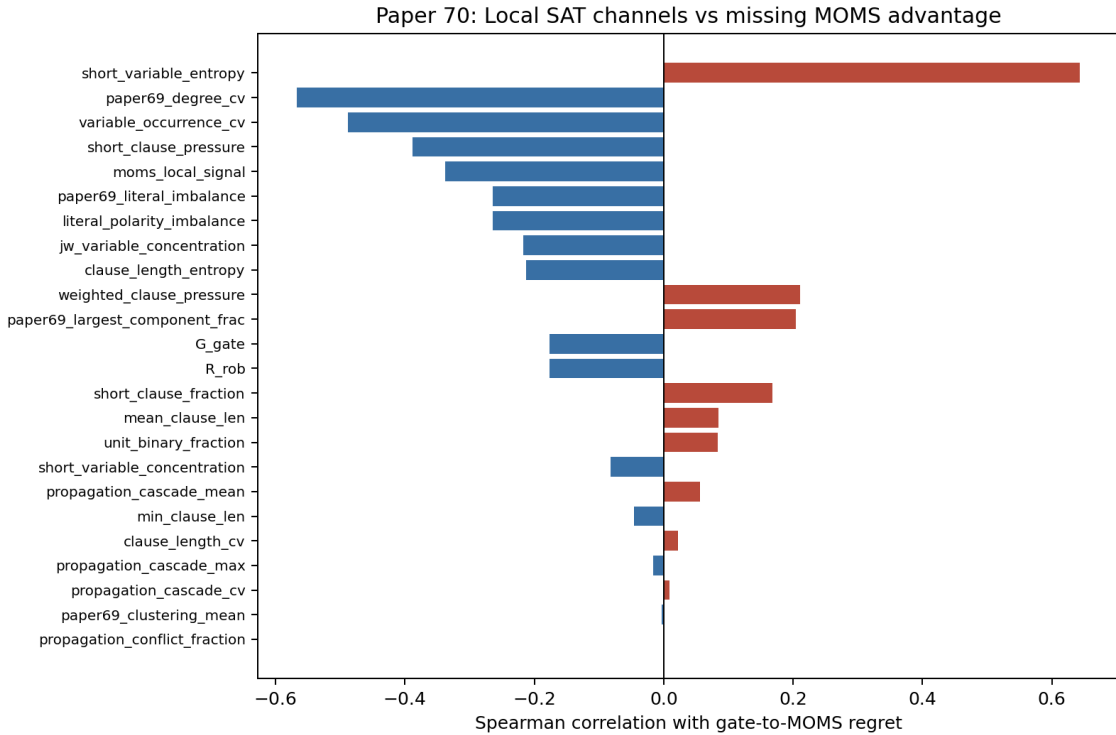


Figure 1: Spearman correlations between local SAT channels and gate-to-MOMS regret. The diagnostic target is not SAT hardness itself, but the specific missing advantage of MOMS over the frozen gate.

5.2 Family-level structure

Table 2 summarizes family-level regret and local-channel means. The largest gate-to-MOMS regret appears in the Dubois and pigeonhole families. The graph-colouring and small random-3SAT families show near-zero regret in this smoke subset.

Family	Mean regret	Median regret	MOMS signal	Short pressure	Mean gate
DIMACS AIM	28.036	29.188	0.341	0.332	0.000
DIMACS Dubois	198.008	229.052	2.432	2.397	0.183
DIMACS pigeonhole	88.137	35.796	4.057	4.000	0.236
Graph-colouring flat30	0.229	0.000	3.044	3.000	0.034
Random 3SAT uf20	0.971	1.632	4.867	4.550	0.816
Random 3SAT uuf50	19.057	21.804	4.495	4.360	0.536

Table 2: Family-level diagnostic summary. The same MOMS-local signal does not imply the same regret across families, which indicates that missing information is not a single scalar pressure but a family-dependent local channel geometry.

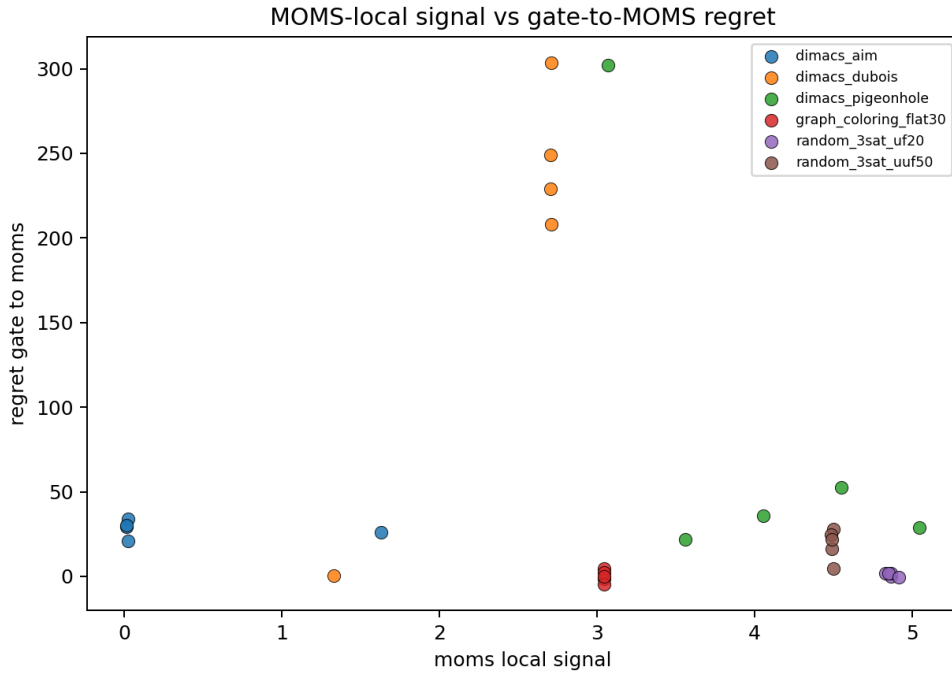


Figure 2: MOMS-local signal versus gate-to-MOMS regret. The relationship is family-structured rather than globally linear. This is why Paper 70 treats the result as missing structural information, not as a new gate formula.

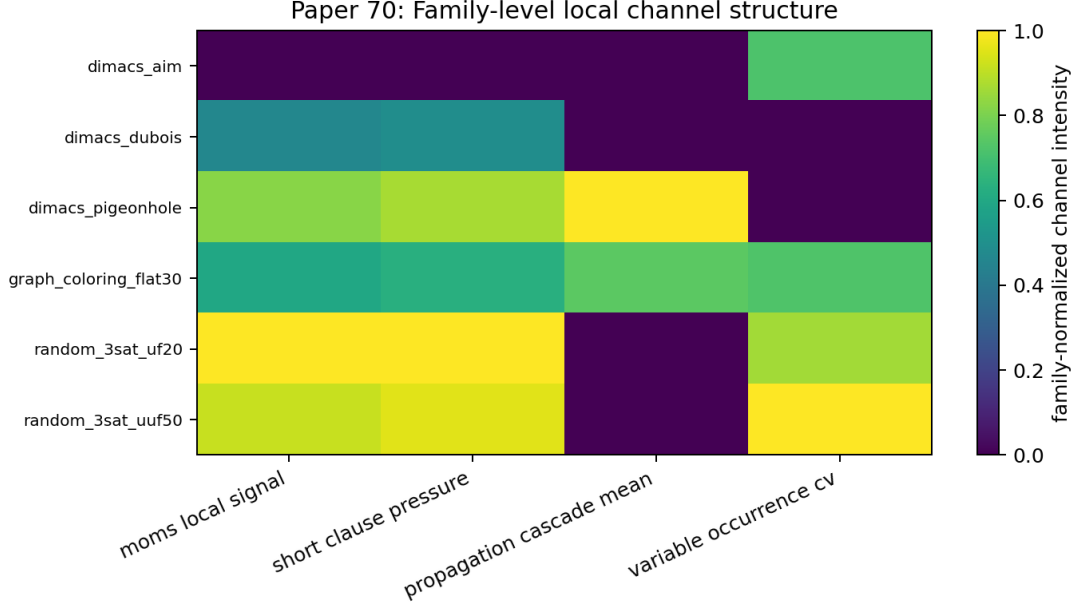


Figure 3: Family-level local-channel intensities. Local SAT structure differs qualitatively across AIM, Dubois, pigeonhole, graph-colouring, and random-3SAT families.

5.3 Propagation proxy

The unit-propagation cascade proxy is weaker in this smoke subset:

$$\rho_S(\text{propagation cascade mean}, r) = 0.056.$$

No initial single-literal propagation conflicts were detected in this diagnostic run. This does not prove that propagation locality is irrelevant. It only shows that this simple root-level propagation proxy does not explain the Paper-69 gate-to-MOMS gap.

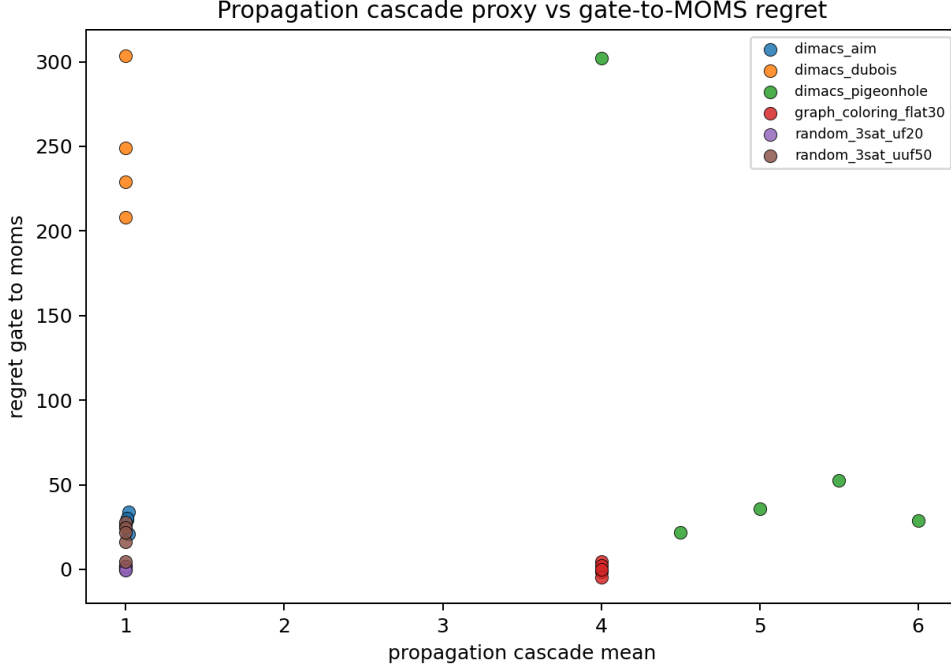


Figure 4: Propagation-cascade proxy versus gate-to-MOMS regret. The proxy is family-structured but weakly correlated with the target in this small smoke subset.

6 Interpretation

The first external SAT Gate Challenge produced a negative result for the frozen gate. Paper 70 sharpens that result:

The current root-level robustness gate does not encode the decision-local shortest-clause/occurrence geometry exploited by MOMS.

This is more informative than saying simply that the gate lost. It identifies a concrete missing channel. The missing channel is local, decision-adjacent, and family-dependent. It is not captured by compressing the instance into a single root-level R_{rob} -style activation coordinate.

The strongest observed channel is shortest-clause variable entropy. This is subtle. The MOMS advantage is not explained by raw short-clause pressure alone. Rather, regret is sensitive to how shortest-clause pressure is distributed across variables. When this pressure is broadly distributed, the root gate has little way to choose locally; MOMS still acts directly on the shortest-clause layer.

This interpretation is consistent with the broader v1.4–v1.7 lesson: some domains cannot be compressed into one global score or gate without losing the information that actually drives decisions.

7 Limitations

The limitations are substantial:

- The analysis is post-hoc and diagnostic.
- The sample contains only 30 family-balanced SATLIB smoke instances.
- The transparent CDCL backend is not CaDiCaL, Kissat, Glucose, or MiniSat.

- The propagation proxy is simple and root-level.
- No solver policy is changed.
- No new gate is proposed or validated.
- Correlations are descriptive and family-structured, not causal proof.

The correct conclusion is therefore not:

“use this channel as the next gate.”

The correct conclusion is:

“the next SAT experiment must preregister a local-channel test.”

8 Reproducibility and Data Policy

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Experiment folder:

experiments/sat_local_structural_channels_paper70/

Run command:

```
python3 analyze_sat_local_channels_paper70.py \
--paper69-dir ../gate_challenge_sat_paper69
```

The script expects the Paper-69 SATLIB smoke data to be staged locally and verified using the Paper-69 SHA256 manifest. Raw SATLIB CNF files are not committed or redistributed. Published Paper-70 outputs contain only derived diagnostics:

- local channel features,
- family-level summaries,
- correlations,
- JSON metadata,
- figures.

The output summary explicitly marks the analysis as:

post_hoc_diagnostic_not_retuning.

9 Conclusion

Paper 70 converts the first external SAT Gate Challenge result into a diagnostic finding. The frozen MAAT v1.7 gate did not beat MOMS because it lacks a local shortest-clause/occurrence channel. This is a missing-information result, not a reason to retroactively change Paper 69.

The practical implication is straightforward. Before any new SAT-specific MAAT policy is tested, the local structural channel must be specified in advance, compared against MOMS and Jeroslow–Wang, and evaluated on external benchmarks under the same falsification discipline used in Paper 68.